

day/date

HW #39

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CS-F

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$A = U \Sigma V^T$$

3×2 3×3 3×2 2×2

$$A^T A$$

$$= \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} = 0$$

$$(2-\lambda)^2 - 1 = 0 \Rightarrow 4 - 4\lambda + \lambda^2 - 1 = 0 \Rightarrow \lambda^2 - 4\lambda + 3 = 0$$

$$\lambda = 3 \quad \lambda = 1$$

$$\begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x = \begin{bmatrix} +1 \\ 1 \end{bmatrix} x_2$$

$$\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x = \begin{bmatrix} -1 \\ 1 \end{bmatrix} x_2$$

$$\begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$V = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$U_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} \sqrt{2} \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$U_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

VICKY

3rd vector :

$$v_1 \cdot v_3 = 0$$

$$\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}^T \begin{bmatrix} a \\ b \end{bmatrix} = 0$$

$$\frac{a+b}{\sqrt{2}} = 0$$

$$v_2 \cdot v_3 = 0$$

$$\begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}^T \begin{bmatrix} a \\ b \end{bmatrix} = 0$$

$$\frac{-a+b}{\sqrt{2}} = 0$$

$$u_1 \cdot u_3 = 0$$

$$\begin{bmatrix} \sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

$$u_2 \cdot u_3 = 0$$

$$\begin{bmatrix} 0 & +1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

$$a\sqrt{2} + \frac{b}{\sqrt{2}} + \frac{c}{\sqrt{2}} = 0 \Rightarrow 2a + b + c = 0$$

$$\frac{b}{\sqrt{2}} - \frac{c}{\sqrt{2}} = 0 \Rightarrow \frac{b}{\sqrt{2}} = \frac{c}{\sqrt{2}}$$

$$a\sqrt{2} + \sqrt{2}c = 0 \quad 2a + 2c = 0$$

$$a + c = 0$$

$$b = c = -a$$

$$u_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

$$2. \sum_{i=1}^3 \sigma_i u_i v_i^T$$

$$A = \begin{bmatrix} \sqrt{2} & 0 & -1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{2} & +1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{2} & +1/\sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} + 1/\sqrt{2} \\ -1/\sqrt{2} + 1/\sqrt{2} \\ 1/\sqrt{2} + 1/\sqrt{2} \end{bmatrix}$$

$$\sigma_1 u_1 v_1^T$$

$$= \sqrt{3} \begin{bmatrix} \sqrt{2} \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$\sigma_2 u_2 v_2^T$$

$$= \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$= \sqrt{3} \begin{bmatrix} 1 & 1 \\ 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ -1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{3} & \sqrt{3} \\ \sqrt{3}/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & \sqrt{3}/2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ -1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ -\frac{1+\sqrt{3}}{2} & \frac{1+\sqrt{3}}{2} \\ \frac{1+\sqrt{3}}{2} & -\frac{1+\sqrt{3}}{2} \end{bmatrix}$$

$$3) \quad A^+ = \frac{1}{\sqrt{3}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \\ 1 & 1/2 & 1/2 \end{bmatrix}$$

$$A^+ = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1/2 & 1/2 \\ 0 & 1/2 & -1/2 \end{bmatrix}$$

$$A^+ = \begin{bmatrix} 1/\sqrt{3} & -\frac{3+\sqrt{3}}{6} & \frac{3+\sqrt{3}}{6} \\ 1/\sqrt{3} & \frac{3+\sqrt{3}}{6} & -\frac{3+\sqrt{3}}{6} \end{bmatrix}$$